Automatic Segmentation and Detection of Pancreatic Ductal Adenocarcinoma (PDAC) in CT

Two-stage pipeline (CT-only pancreas localizer + nnU-Net Residual Encoder U-Net) trained and evaluated on the full PANORAMA dataset (2,238 contrast-enhanced CT studies)

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Document: Final project report: dataset, method, architecture, workflow, results and limitations

Compute: Kaggle notebooks, NVIDIA Tesla T4 (16 GB); durable storage in a private Hugging Face repository

Date of final run: 21 October 2026

Status: training complete (300 epochs); single locked evaluation on a held-out test set (338 studies)

Summary

Headline result (held-out test set, never used for training or tuning). On the 72 test studies with manually drawn tumour masks, the model reached a mean tumour Dice of **0.635** (95% CI 0.559-0.705) and a median of **0.776**; it found 87.5% of tumours. As a PDAC-vs-non-PDAC classifier over all 338 test studies it reached an AUROC of **0.952** (95% CI 0.926-0.975), with sensitivity 93.1% and specificity 84.0% at the operating point fixed on validation data.

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1. Problem and objectives

Pancreatic ductal adenocarcinoma (PDAC) is the most common pancreatic cancer and has one of the lowest survival rates of any cancer, partly because it is often detected late. On contrast-enhanced CT the tumour is small relative to the scan (median about 11 mL, roughly 0.02% of the volume) and is only about 17 HU darker than normal pancreas, so it is hard to see and hard to segment automatically.

Objectives of this project:

1. Build a fully automatic pipeline that takes a raw abdominal CT and outputs a 3D mask of the PDAC lesion (and the pancreas).
2. Use the **entire** public PANORAMA dataset (2,238 studies, 194 GB), not a subset.
3. Evaluate honestly: patient-level split, no information from the test set used for any decision, rules for model selection written down before results were seen (pre-registration).
4. Report both segmentation quality (Dice, HD95) and patient-level detection (sensitivity, specificity, AUROC), with 95% confidence intervals.
5. Make the whole pipeline resumable on free cloud GPUs with hard session limits.

2. Dataset: PANORAMA

PANORAMA (*Pancreatic Cancer Diagnosis: Radiologists Meet AI*) is a public challenge dataset from Radboud University Medical Center (Alves et al., 2024). The public training/development cohort was used, as mirrored on Hugging Face (`MedOtter/PANORAMA`). The challenge's own hidden validation/test sets are not public and were not used. Licence: CC BY-NC 4.0.

2.1 Size and contents

Item	Value
CT studies / patients	2,238 studies from 2,224 patients (11 patients have more than one study)
Imaging	Contrast-enhanced portal-venous abdominal CT, NIfTI (.nii.gz)
Total image size	194.4 GB (median 68.6 MB, largest 889.7 MB per study)
Voxel spacing	in-plane about 0.6-0.9 mm; slice thickness 0.5-5.0 mm; 75-1,141 slices
Diagnosis	676 PDAC, 1,562 non-PDAC studies
Manual (gold) masks	482 studies (479 PDAC, 3 non-PDAC)
Automatic (silver) masks	1,756 studies (197 PDAC, 1,559 non-PDAC)

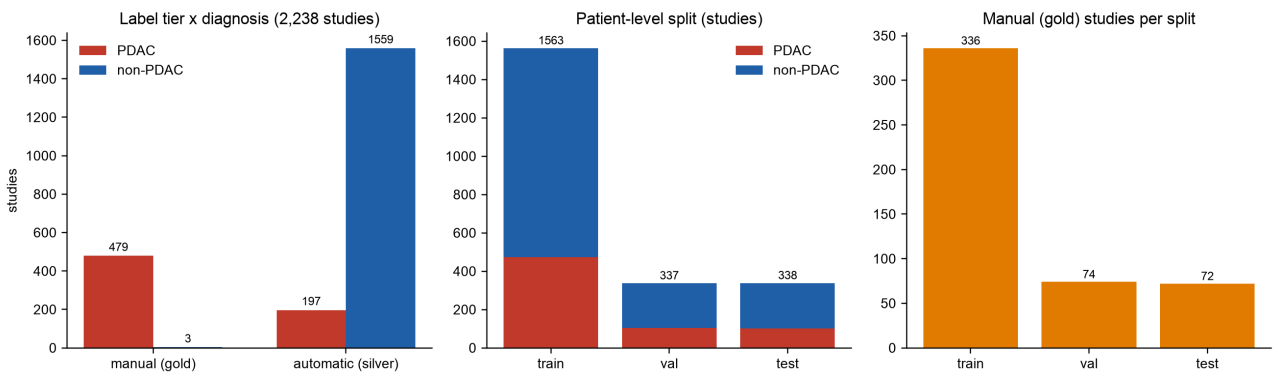


Figure 2.1. Studies by label tier and diagnosis, the locked patient-level split, and the number of manually annotated studies per split.

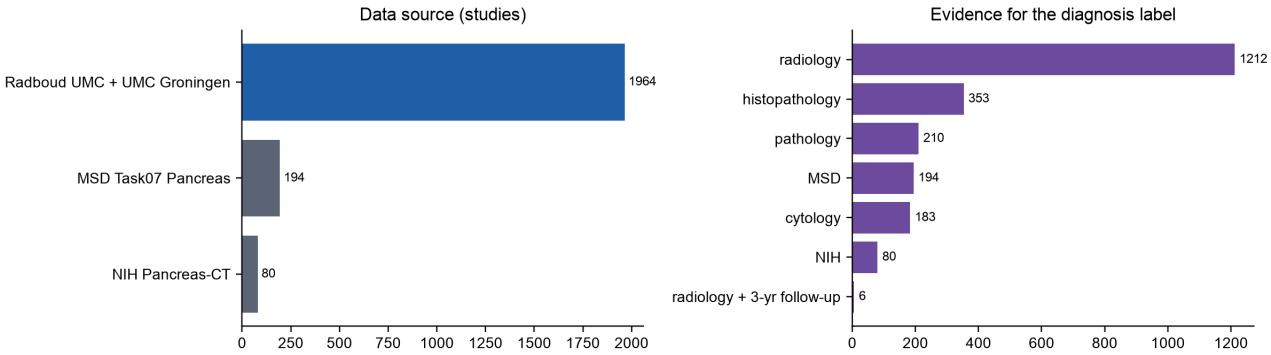


Figure 2.2. Data sources and the evidence behind the diagnosis label (from clinical_information.csv).

2.2 Label definitions

Each mask has 7 values: 0 background, **1 PDAC lesion**, 2 veins, 3 arteries, 4 pancreas parenchyma, 5 pancreatic duct, 6 common bile duct. In **manual** cases the lesion was drawn by trained investigators supervised by an expert radiologist; the anatomy labels (2-6) are produced by an algorithm in every case. In **automatic** cases the lesion is also algorithm-generated. Therefore:

- Tumour Dice against **manual** masks is the only gold-standard segmentation measure.
- Tumour Dice against automatic masks is reported separately as a 'silver' measure.
- Pancreas Dice is agreement with an algorithm, not with a radiologist (marked with † below).

For this project the 7 labels were mapped to 3 training classes: **0 background**, **1 pancreas** (labels 4 and 5), **2 PDAC** (label 1). Vessels and bile duct were treated as background.

2.3 Hazards identified before modelling

- **Almost no gold negatives:** only 3 manual non-PDAC studies exist, so false positives on healthy pancreas can only be measured on automatic-tier non-PDAC studies (diagnosis label is still clinical).
- **Repeat studies:** 11 patients have several studies, so the split must be by patient.
- **Heterogeneous geometry:** mixed orientations (RAS/LPS) and slice thickness, so everything is reoriented and resampled.
- **The Hugging Face 'viewer' version is 2D PNG previews only**; all work used the 3D NIfTI files.
- MSD and NIH cases are embedded in the dataset; results must not be compared directly with benchmarks on those datasets.

3. Approach overview

A single 3D network cannot process a whole abdominal CT at full resolution on a 16 GB GPU, and the tumour occupies a tiny fraction of the scan. The pipeline therefore works in two stages, which is the standard design for pancreas tumour segmentation:

1. **Stage 1 - pancreas localizer.** A small 3D CNN looks at a coarse (4 mm) version of the whole CT and predicts the 3D centre of the pancreas. It uses only the CT, never the labels.
2. **ROI crop.** A fixed box of 96 x 224 x 288 voxels (192 x 224 x 288 mm) is cut around the predicted centre at 1 x 1 x 2 mm resolution. This box is large enough to contain the whole pancreas even when the localizer is off by several centimetres.
3. **Stage 2 - segmentation.** nnU-Net with a Residual Encoder U-Net (ResEnc-M) segments pancreas and PDAC inside the crop.
4. **Post-processing and decision.** The PDAC probability map is thresholded, small components are removed, and a study is called PDAC-positive if any lesion remains. All settings were chosen on validation data only.

Why these choices: nnU-Net is the strongest general baseline in 3D medical segmentation and the Residual Encoder variant was shown in the 2024 nnU-Net benchmark study (Isensee et al., MICCAI 2024) to outperform many newer transformer and Mamba architectures. Training only on manual masks would give the network almost no healthy pancreas to learn from, so all 1,557 training studies (manual and automatic, PDAC and non-PDAC) were used for Stage 2.

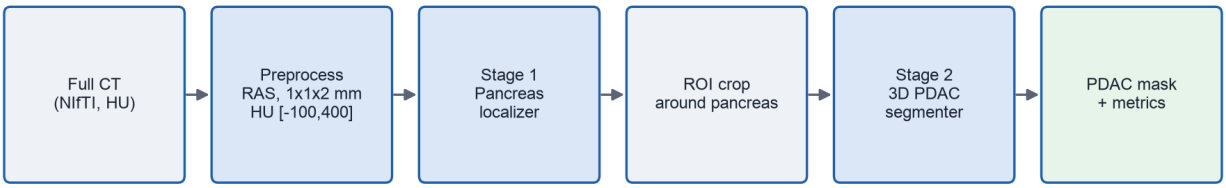


Figure 3.1. Two-stage cascade used in this project.

3.1 Compute and storage design

Kaggle gives a free Tesla T4 GPU but limits a session to about 12 hours and 20 GB of persistent disk, while the dataset is 194 GB. The pipeline was therefore designed to **stream** data case by case and to keep every intermediate result in a private Hugging Face repository that acts as durable storage. Each phase writes a 'done' marker; re-running the notebook skips finished phases and resumes unfinished ones. The same notebook runs on Kaggle, Google Colab or a local machine.

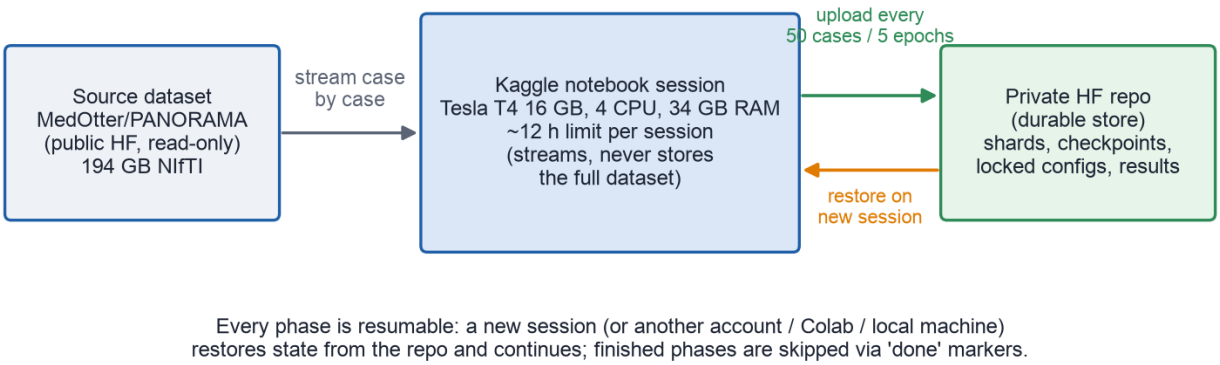


Figure 3.2. Storage and resumability design.

4. Step-by-step workflow

The notebook has 13 cells. Each cell is idempotent: it checks the repository first and only does work that has not already been stored.

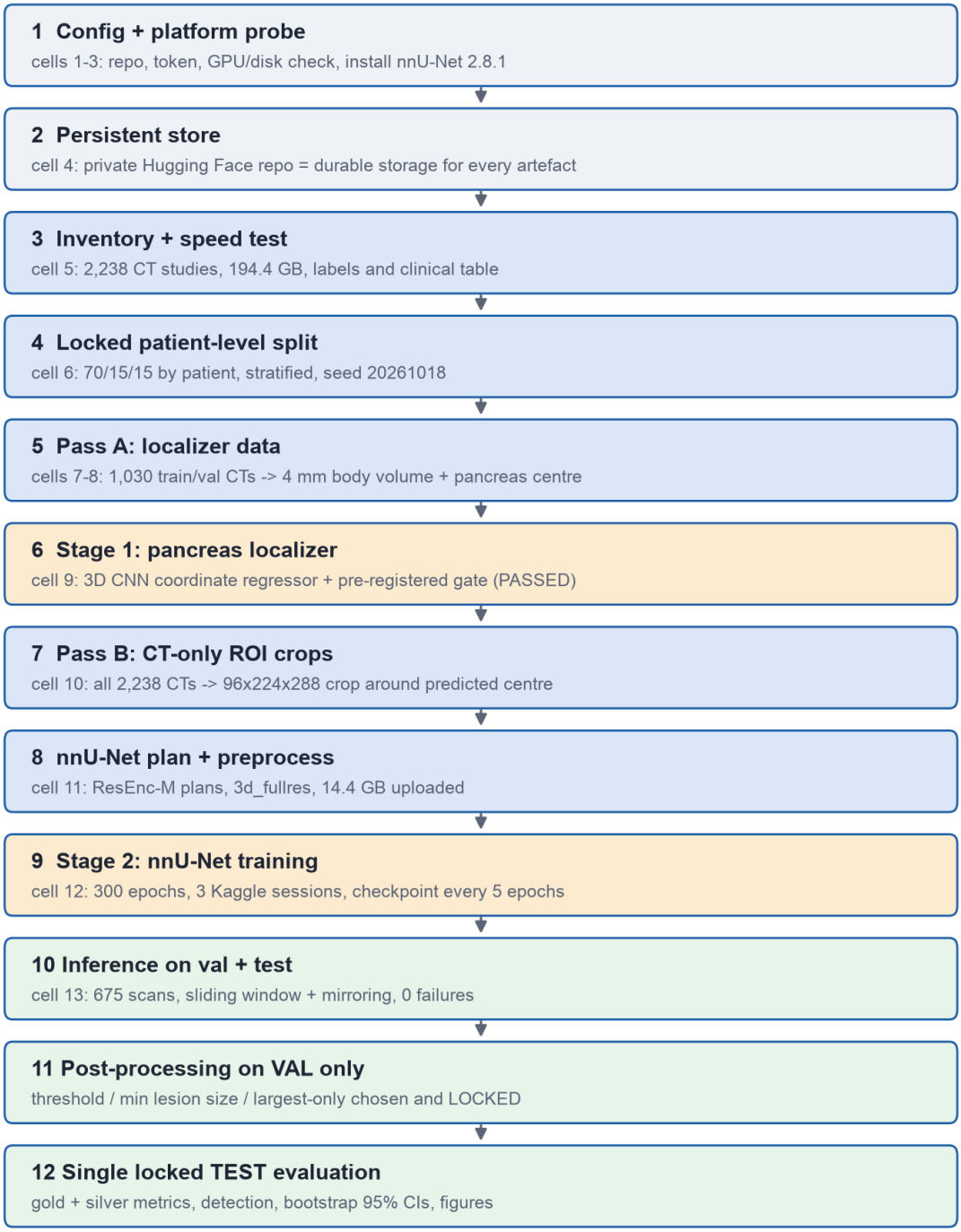


Figure 4.1. End-to-end workflow. Orange = model training stages, green = evaluation.

Cell	Purpose	Key outputs from the actual run
1	Configuration (repositories, seed 20261018)	only cell edited by hand
2	Platform probe	Kaggle, Python 3.13, 4 CPU, 33.7 GB RAM, Tesla T4 x2 (one used)
3	Install + verify nnU-Net	nnunetv2 2.8.1, torch 2.11.0+cu128

Cell	Purpose	Key outputs from the actual run
4	Private HF store + write/read test	round-trip OK; phase markers
5	Inventory + download speed test	2,238 cases, 0 missing images/labels, about 39 MB/s
6	Locked patient-level split	train 1,563 / val 337 / test 338 studies; no patient overlap
7	Preprocessing + streaming engine	shards of 50 cases, retries, session time guard
8	Pass A: localizer training data	1,030 train/val studies, 0 failures, 7.8 s/case
9	Stage 1 localizer + gate	best epoch 38; pre-registered gate PASSED
10	Pass B: CT-only ROI crops for all studies	2,238 crops, 17.0 GB, 11.2 s/case
11	nnU-Net dataset, plans, preprocessing	ResEnc-M plans, 14.4 GB preprocessed, 8 parts
12	nnU-Net training (resumable)	300 epochs over 3 sessions, about 24.9 GPU-hours
13	Inference, selection on val, locked test evaluation	675 scans, 0 failures; results in Section 9

5. Data split and preprocessing

5.1 Patient-level split (cell 6)

Patients, not studies, were assigned to train / validation / test (70 / 15 / 15%), stratified by diagnosis and label tier (scikit-learn `train_test_split`, seed 20261018). The split was stored in the repository and reused unchanged in every later session. A check confirmed that no patient appears in two splits.

Split	Patients	Studies	PDAC	non-PDAC	Manual (gold)	Automatic
Train	1,556	1,563	472	1,091	336	1,227
Validation	334	337	103	234	74	263
Test	334	338	101	237	72	266

The test set was not opened for any purpose until the single final evaluation in cell 13. Its 72 manual studies are all PDAC; its 237 non-PDAC studies come from the automatic tier.

5.2 CT preprocessing (cell 7)

Step	Setting	Reason
Reorientation	RAS (SimpleITK DICOMOrient)	datasets mix RAS and LPS
Resampling	1 x 1 x 2 mm (x, y, z); linear for CT, nearest for masks	uniform voxel size
Intensity window	-100 to 400 HU, scaled to [0, 1] (localizer)	keeps 99.99% of tumour voxels
Body crop	voxels above -500 HU plus a 16 mm margin	removes air; never cut tumour
Coarse volume	average pooling 2 x 4 x 4 to 4 mm isotropic	whole-body input for Stage 1

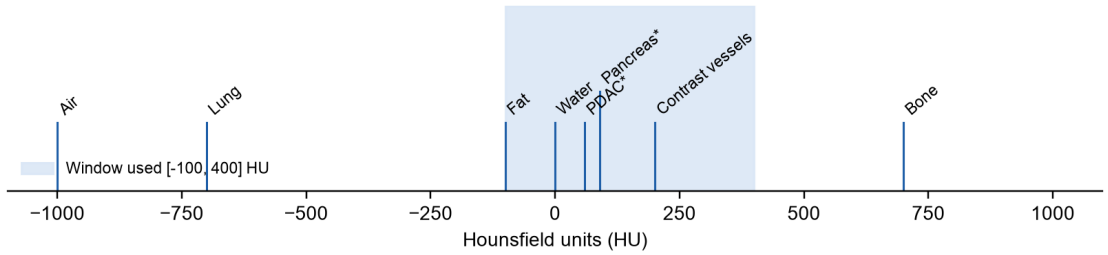


Figure 5.1. HU window used. PDAC and pancreas differ by only about 17 HU on average, which is why the task is hard.

Pass A checks: mask volume after resampling matched the original (median ratio 1.000; none outside 0.8-1.25), and no tumour voxel was lost by the body crop. Pancreas extent was at most 176 x 133 x 228 mm (z, y, x), which fixed the ROI size.

6. Stage 1: pancreas localizer

6.1 Model

A light 3D CNN regresses the pancreas centre from the coarse 4 mm whole-body CT. Three extra input channels hold normalised z, y, x coordinates ('CoordConv') so that the network can reason about absolute position. Training data: 836 training and 194 validation studies from Pass A (all manual train/val studies plus 620 automatic ones, 25% PDAC).

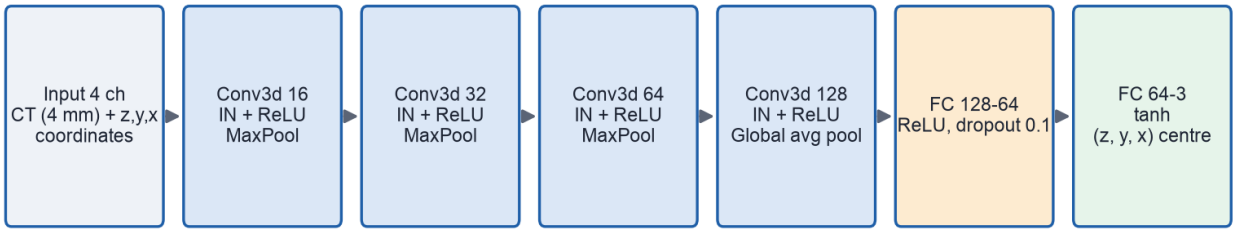


Figure 6.1. Localizer architecture (300,483 parameters).

Hyper-parameter	Value
Target	pancreas+PDAC centre of mass (labels 1, 4, 5), normalised to [-1, 1]
Loss	Smooth L1 / Huber, beta 0.1
Optimiser	AdamW, learning rate 2e-4, weight decay 1e-4, cosine schedule, 40 epochs
Augmentation	intensity scaling and shift, Gaussian noise
Validation	every 2 epochs; checkpoint with lowest mean error kept (epoch 38)

6.2 Pre-registered gate

Before training, pass/fail criteria were fixed. Stage 2 was allowed to start only if all passed:

Criterion (fixed in advance)	Threshold	Result	
Manual val: median centre error	<= 25 mm	15.1 mm	PASS
Manual val: within 40 mm	>= 90%	97.3%	PASS
Automatic val: median centre error	<= 30 mm	17.9 mm	PASS
Automatic val: within 40 mm	>= 85%	98.3%	PASS
ROI contains the pancreas (all val)	>= 95%	100%	PASS

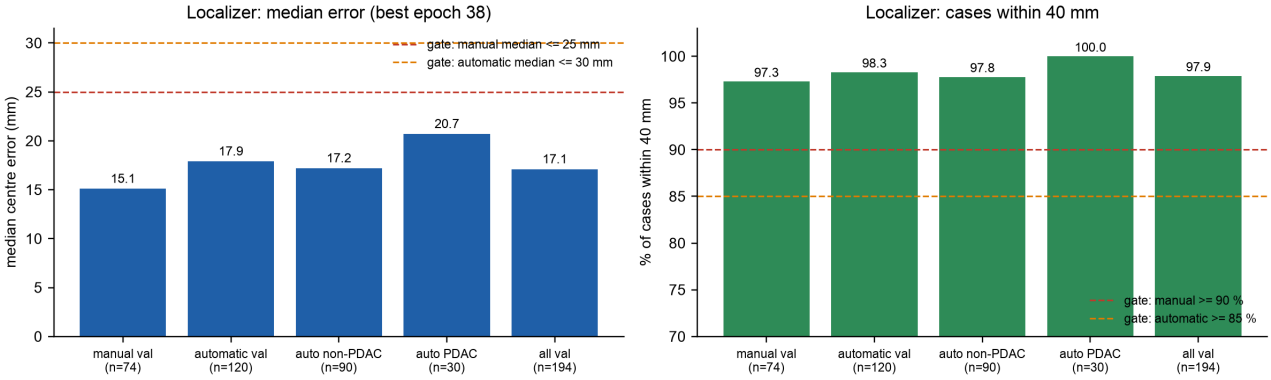


Figure 6.2. Localizer error on the validation set by subgroup. Non-PDAC scans were checked separately because the manual training data are almost all PDAC.

6.3 ROI crops for every study (Pass B, cell 10)

The locked localizer predicted a centre for all 2,238 studies from CT alone, and a 96 x 224 x 288 voxel box was cropped around it. The masks were cropped with the same box only to build training targets and to *measure* what the crop kept.

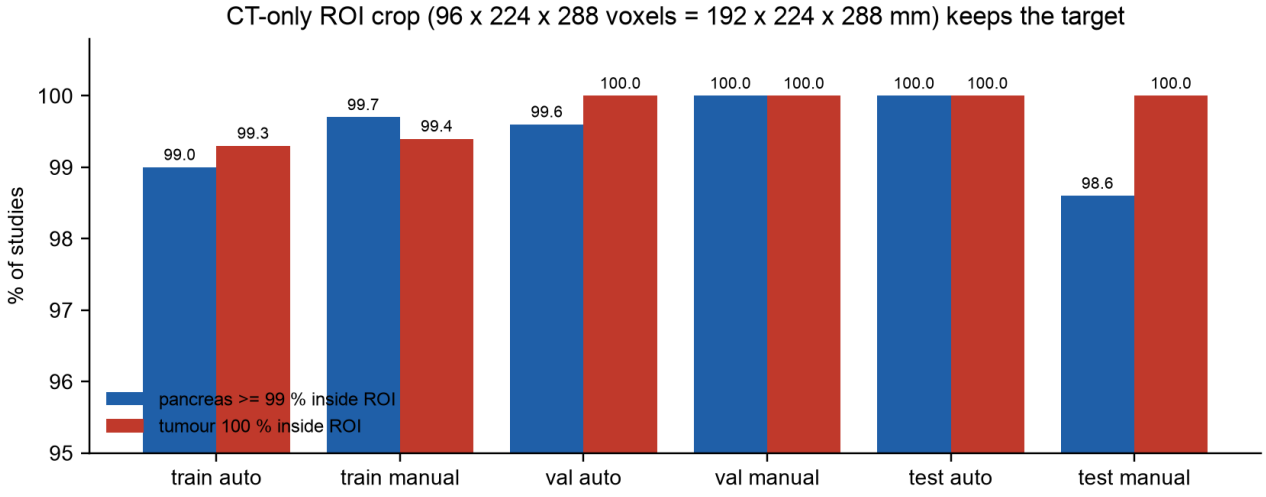


Figure 6.3. The CT-only crop kept the whole tumour in 100% of validation and test PDAC studies.

Six training studies in which the crop missed most of the target (tumour or pancreas under 50% inside) were excluded from training only: 100028, 100118, 100852, 100939, 101303, 101379. No validation or test study was excluded.

7. Stage 2: nnU-Net segmentation model

7.1 Self-configuration (cell 11)

nnU-Net v2 (Isensee et al., Nature Methods 2021) automatically derives preprocessing, network topology and training settings from a 'fingerprint' of the dataset. The `nnUNetPlannerResEncM` planner (Residual Encoder, medium preset for about 9-11 GB of GPU memory) was used with the 3d_fullres configuration. The locked split was given to nnU-Net as fold 0 (train = training split, validation = validation split).

Planned setting	Value
Input	1 channel (CT), crops of 96 x 224 x 288 voxels
Target spacing	2.0 x 1.0 x 1.0 mm (z, y, x)
Patch size / batch size	64 x 128 x 192 voxels / 2
Normalisation	CTNormalization: clip to foreground 0.5-99.5 percentile (-57 to 186 HU), z-score with mean 77.7, s.d. 43.7
Network	ResidualEncoderUNet, 6 stages, features 32-64-128-256-320-320
Residual blocks per encoder stage	1, 3, 4, 6, 6, 6
Normalisation / activation	InstanceNorm3d / LeakyReLU
Output classes	3 (background, pancreas, PDAC), softmax

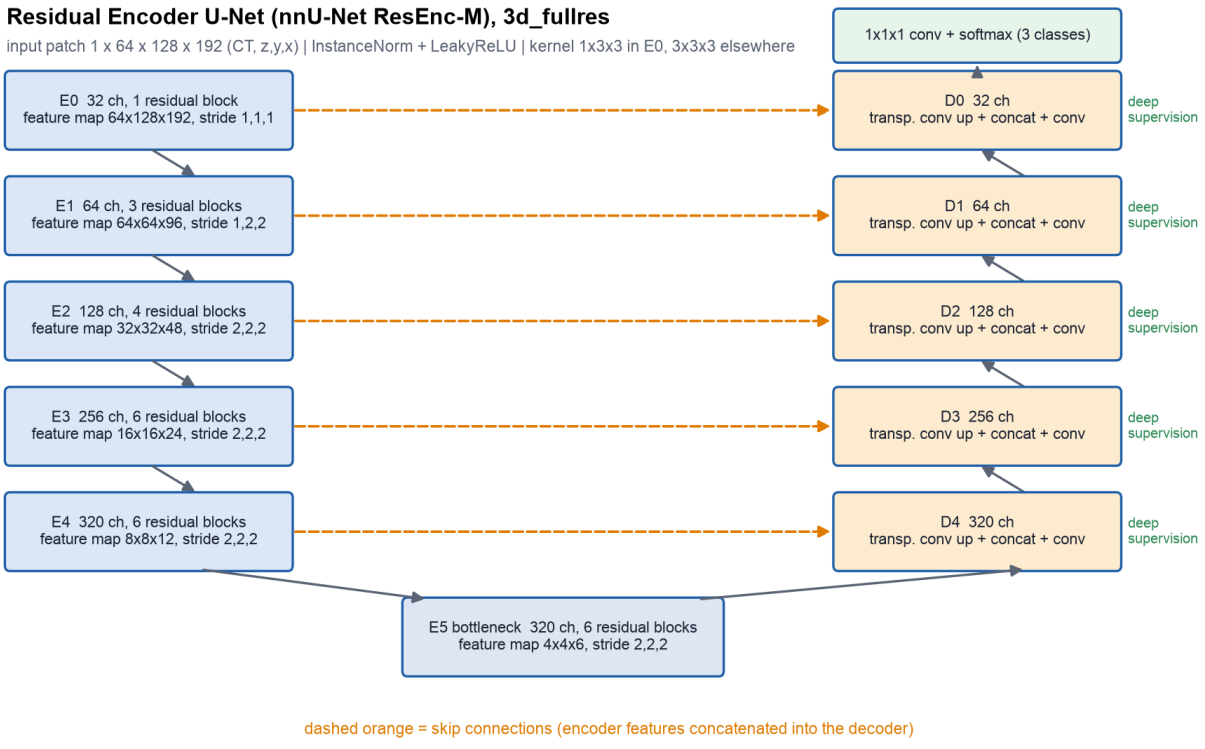


Figure 7.1. Architecture of the segmentation network as planned by nnU-Net for this dataset. Each encoder stage halves the resolution; the decoder mirrors it, and every decoder level has an auxiliary output (deep supervision).

7.2 How a residual encoder U-Net works

A U-Net has an **encoder** that repeatedly convolves and downsamples the image to capture context (what organ is this, where is the tumour likely), and a **decoder** that upsamples back to full resolution to place boundaries precisely. **Skip connections** pass fine detail from each encoder level directly to the matching decoder level. In the **residual encoder**, every block adds its input to its output (x + F(x)); this keeps gradients healthy in a deep encoder (26 residual blocks here) and lets the network be much deeper than a plain U-Net at the same memory cost. **Deep supervision** applies the loss at several decoder resolutions, which speeds up and stabilises training.

7.3 Training recipe (cell 12)

Setting	Value	Note
Loss	Dice + cross-entropy, with deep supervision	nnU-Net default
Optimiser	SGD, momentum 0.99, Nesterov, weight decay 3e-5	nnU-Net default
Learning rate	0.01, polynomial decay (exponent 0.9) to 0	Figure 8.1
Epochs	300 x 250 iterations (75,000 iterations)	reduced from the default 1,000 to fit the free GPU budget
Foreground oversampling	33% of patches forced to contain foreground	nnU-Net default
Augmentation	rotation, scaling, Gaussian noise and blur, brightness, contrast, low-resolution simulation, gamma, mirroring	nnU-Net default (on the fly)
Model selection	final checkpoint (epoch 300)	fixed before evaluation
Built-in validation	skipped; replaced by our own pre-registered evaluation	cell 13

A custom trainer (`nnUNetTrainer_PANORAMA`) changes nothing in the recipe above. It only (a) sets the number of epochs, (b) uploads the checkpoint, progress plot and logs to the repository every 5 epochs, and (c) stops cleanly shortly before the Kaggle session limit so the next session can resume exactly where it stopped.

8. Training history

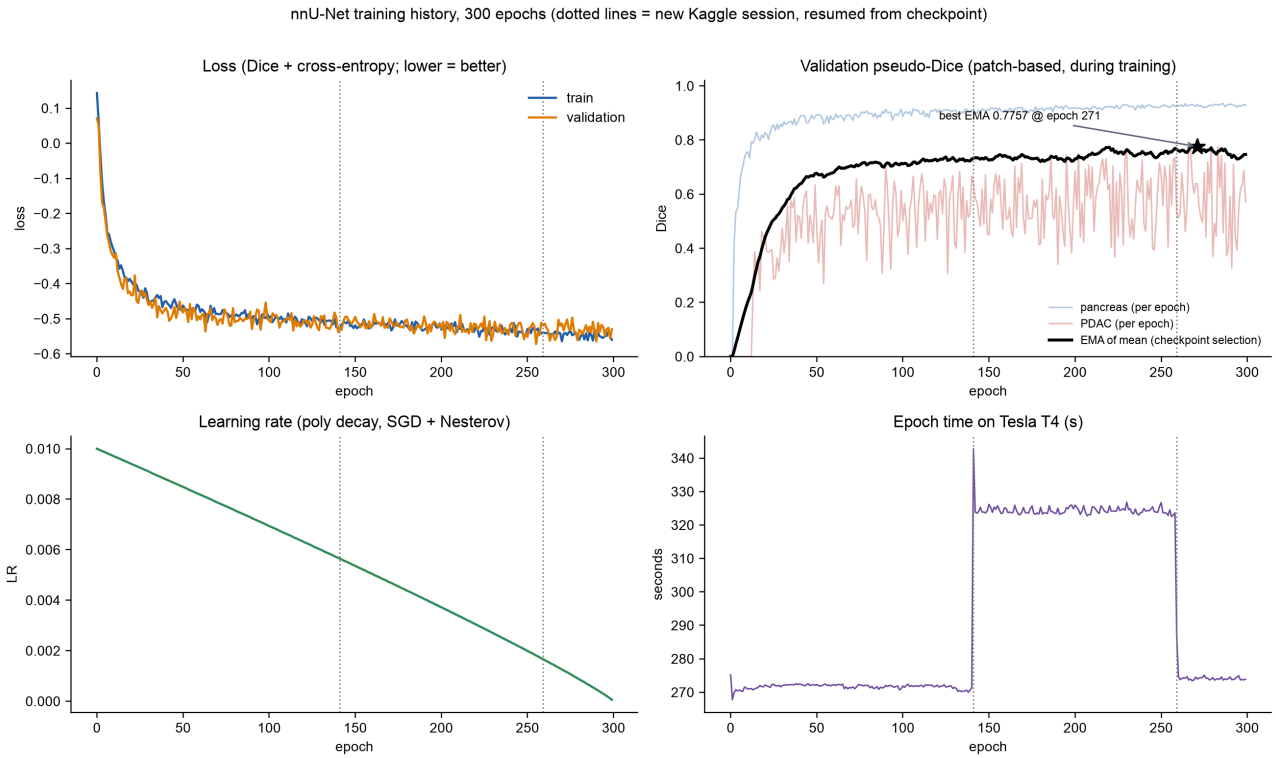


Figure 8.1. Per-epoch values from the nnU-Net logs. Pseudo-Dice is computed on random validation patches during training (tumour-biased), so it is a progress indicator, not the final result. Epoch time changed because Kaggle assigned machines of different speed in each session.

Quantity	Value
Epochs completed	300 / 300 (three sessions: epochs 0-140, 141-258, 259-299)
Best EMA pseudo-Dice (mean of pancreas and PDAC)	0.7757 at epoch 271
Mean pseudo-Dice over the last 20 epochs	pancreas 0.929, PDAC 0.545
Final train / validation loss	-0.5599 / -0.5296
Mean epoch time	293 s on a Tesla T4
Total training time	about 24.9 GPU-hours

Training and validation loss stay close together throughout, so there is no sign of over-fitting. The PDAC pseudo-Dice fluctuates strongly between epochs because each epoch evaluates only a few small tumours; the moving average (EMA) rises steadily, with the largest gains in the low-learning-rate phase after epoch 200.

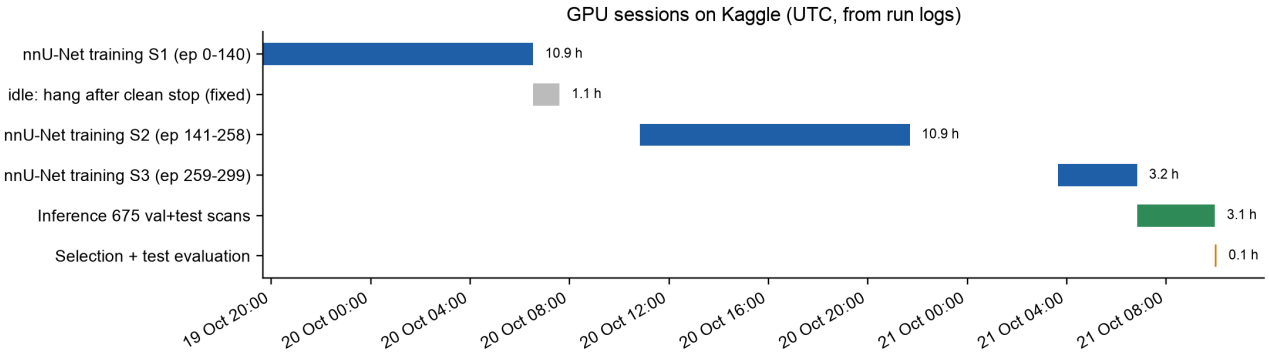


Figure 8.2. GPU sessions (UTC). The grey bar is GPU time lost to a process hang after the first clean stop; the cause was found and fixed (Section 12).

9. Evaluation protocol

To make the test result trustworthy, every rule was written into the repository (`eval/protocol.json`) **before** any metric was computed, and the notebook refuses to run if the protocol is changed afterwards.

1. **Inference.** nnU-Net sliding-window prediction (50% tile overlap, Gaussian weighting, test-time mirroring) on the ROI crops of all 337 validation and 338 test studies with checkpoint_final. 675 studies, 0 failures, 3.1 h.
2. **Candidate post-processing (fixed grid).** PDAC probability threshold in {0.2, 0.3, ..., 0.8}; minimum lesion size in {0, 100, 250, 500, 1000} mm^3; keep all lesions or only the largest (70 combinations; 26-connected components).
3. **Selection rule (validation only).** Maximise the mean tumour Dice on the 73 manual validation PDAC studies, subject to a false-positive rate of at most 15% on the 234 validation non-PDAC studies.
4. **Lock.** The selected configuration was stored (`eval/selected_config.json`) and then applied once to the test set.
5. **Uncertainty.** 95% confidence intervals from 2,000 bootstrap resamples over *patients*.

Metric	Definition
Dice	2 x overlap / (predicted + reference) tumour voxels (1 = perfect, 0 = no overlap); a missed tumour counts as 0
HD95	95th percentile of surface distances in mm (computed when a lesion is predicted)
Detection sensitivity	fraction of reference tumours overlapped by the prediction
Patient sensitivity / specificity	study called PDAC if any lesion survives post-processing
AUROC / AP	patient score = maximum PDAC probability in the ROI

Locked configuration

Selected on validation: threshold 0.2, minimum lesion size 100 mm^3, keep all lesions. Validation mean tumour Dice 0.624 (73 gold cases), validation false-positive rate 14.96% (234 non-PDAC studies; constraint 15% met).

10. Results on the held-out test set

Metric	n	Value	95% CI
Tumour Dice, mean (gold)	72	**0.635**	0.559 - 0.705
Tumour Dice, median (gold)	72	0.776	0.659 - 0.814
Tumour detection sensitivity (gold)	72	0.875 (63/72)	0.792 - 0.944
Tumour HD95, median (gold, lesion predicted)	65	4.47 mm	3.61 - 5.39 mm
Pancreas Dice, mean (gold) †	72	0.925	0.909 - 0.937
Tumour Dice, mean (silver labels)	29	0.842	0.806 - 0.873
Patient sensitivity (all PDAC)	101	0.931 (94/101)	0.881 - 0.980
Patient specificity (non-PDAC)	237	0.840 (199/237)	0.793 - 0.886
Patient AUROC	338	**0.952**	0.926 - 0.975
Patient average precision	338	0.871	0.805 - 0.927

† Reference pancreas masks are algorithm-generated in PANORAMA, also for manual cases.

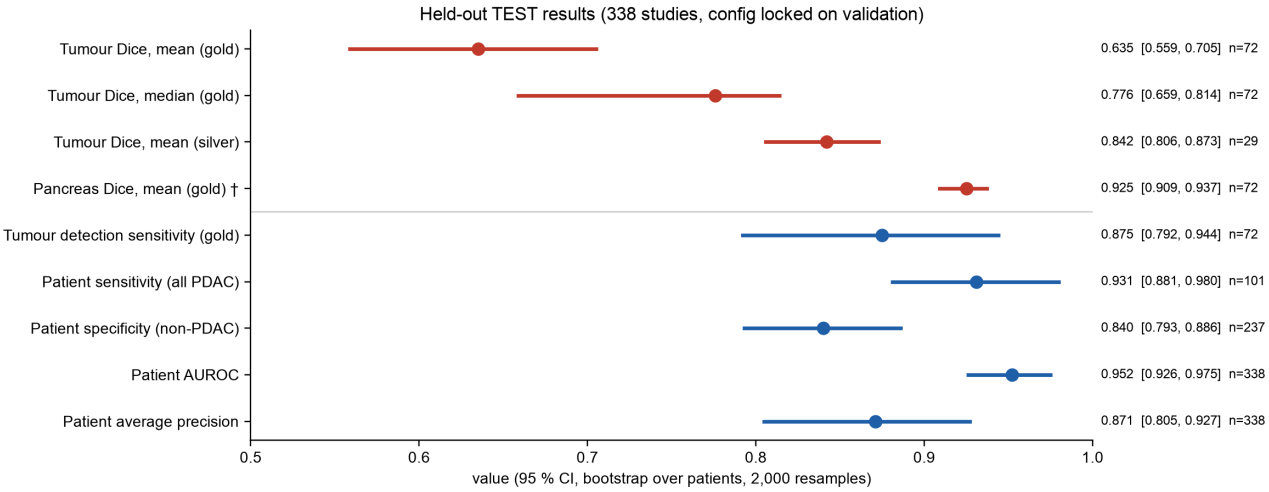


Figure 10.1. Test metrics with 95% bootstrap confidence intervals (red = segmentation, blue = detection).

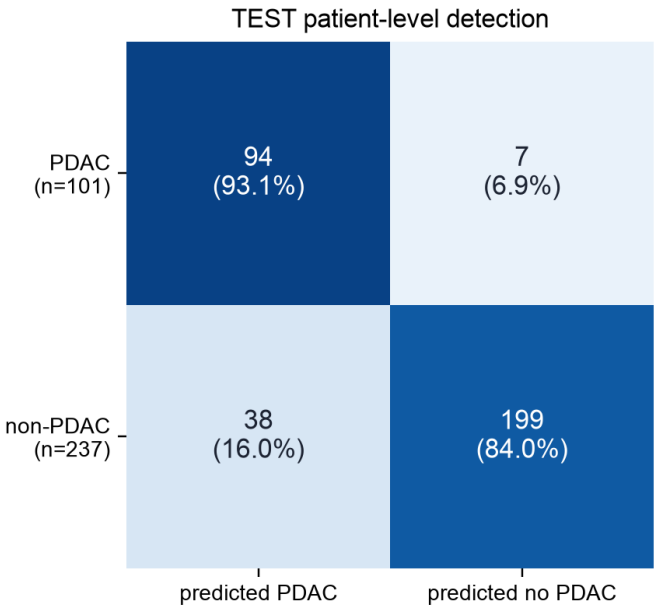


Figure 10.2. Patient-level confusion matrix at the locked operating point.

10.1 Interpretation

- **When a tumour is found, its outline is usually accurate:** the median Dice is 0.776 and the median boundary error (HD95) is 4.5 mm, about the size of 2-4 voxels.
- **The mean Dice (0.635) is lower than the median** because 9 of the 72 gold tumours were missed completely and each counts as Dice 0. Missed tumours are the main weakness.
- **Detection is strong:** AUROC 0.952 separates PDAC from non-PDAC studies well. At the chosen operating point 94 of 101 cancers were flagged, while 38 of 237 non-PDAC studies received a false lesion (specificity 84%). The low threshold (0.2) favours finding cancers over avoiding false alarms, which is the clinically preferred direction for a screening aid.
- **Silver Dice (0.842) is higher than gold Dice** because automatic labels were produced by an algorithm with a similar style to the training targets; it should not be read as real accuracy.
- **Validation and test agree** (validation gold mean Dice 0.624 vs test 0.635; validation FP rate 15.0% vs test 16.0%), which suggests the selection did not over-fit the validation set.

10.2 Additional figures from the run

Cell 13 also saved, in the Kaggle output folder and in the repository under `eval/figures/`: per-case Dice distributions, Dice versus tumour size, predicted versus true tumour volume, ROC and precision-recall curves, example CT slices with reference and predicted outlines (best / median / worst case), and the validation selection curves. They can be appended to this report.

11. Limitations and threats to validity

1. **Internal test set.** The test set is a held-out part of the public PANORAMA cohort (same hospitals and scanners). It is not the official PANORAMA hidden test set, and results are not directly comparable to the challenge leaderboard.
2. **Evaluation space.** Metrics are computed inside the CT-only ROI crop at 1 x 1 x 2 mm, not at the original scan resolution. Every test tumour lay fully inside its crop, so no tumour was lost, but voxel-size differences can shift Dice slightly.
3. **Single model, single split.** One fold, 300 epochs instead of nnU-Net's default 1,000, no cross-validation ensemble. Both changes typically cost a few Dice points.
4. **Label noise.** 78% of training studies have algorithm-generated (silver) masks, and all pancreas references are algorithm-generated.
5. **Few gold negatives.** Specificity is measured on non-PDAC studies whose diagnosis is clinical but whose masks are automatic.
6. **Threshold at the edge of the grid.** The selected threshold (0.2) is the lowest value in the pre-registered grid; a lower one might have been chosen with a wider grid. It was deliberately not changed after the test results were seen.
7. **Repeat studies.** The 338 test studies come from 334 patients; confidence intervals resample patients, not studies, to account for this.

12. Engineering log: problems met and how they were solved

Problem	Cause	Fix
Dataset larger than disk (194 GB vs 20 GB)	Kaggle storage limit	stream case by case; store compact results in shards of 50 in a private repo
Sessions end after about 12 h	Kaggle limit	every phase resumable; checkpoints every 5 epochs; clean stop at 10.9 h
Trainer crashed at start (KeyError 'args')	nnU-Net 2.8 constructs trainers with keyword arguments	explicit constructor signature
Run killed at 12 h after a clean stop	data-loader worker processes kept the output pipe open after the trainer exited	kill worker processes before exit; process-group kill; watchdog at 11.25 h. No training lost (resumed at epoch 141)
Data-loader warning at epoch 245	one augmentation worker died once	no effect: training continued normally (all later epochs logged)
Different epoch speed per session	Kaggle hardware variation (272 s vs 324 s)	none needed; schedule depends on epochs, not time

13. Reproducibility

Item	Value
Code	one Kaggle notebook, 13 cells, platform-independent (Kaggle / Colab / local)
Software	Python 3.13, PyTorch 2.11.0 (CUDA 12.8), nnunetv2 2.8.1, dynamic_network_architectures 0.4.4, batchgeneratorsv2 0.3.6, SimpleITK 2.5.6, scikit-learn 1.6.1, numpy 2.1.3
Hardware	NVIDIA Tesla T4 16 GB, 4 vCPU, 33.7 GB RAM
Random seed	20261018 (split, cohort sampling, bootstrap)
Locked artefacts in the repository	meta/split.csv, passA/cohort.csv, passB/config.json, localizer/best.pt, nnunet_train/config.json, eval/protocol.json, eval/selected_config.json
Results in the repository	nnunet_train/fold_0/checkpoint_final.pth, eval/test_results.json, eval/test_per_case.csv, eval/val_grid.csv, eval/figures/*.png
Compute used	about 24.9 GPU-hours training + 3.1 h inference, plus CPU sessions for data preparation

To reproduce: open the notebook, put a write token for the storage repository in cell 1, select a GPU, and run all cells. Finished phases are restored from the repository, so a full re-run of the evaluation takes minutes, and a run from scratch repeats every step deterministically up to GPU non-determinism.

13.1 Provenance of results

All experiments were run by the author on Kaggle, and every number in this report is copied from the run outputs; nothing was estimated or edited by hand.

14. Conclusion and next steps

A complete, leakage-free pipeline was built and run on the full PANORAMA dataset with free cloud GPUs. On held-out data the model detects PDAC with an AUROC of 0.952 and segments detected tumours with a median Dice of 0.776; the main weakness is a set of small or subtle tumours that are missed entirely.

Most promising improvements:

- Train for the full 1,000 epochs and ensemble 5 cross-validation folds (standard nnU-Net practice).
- Add a patient-level classifier on top of the segmentation (lesion volume, maximum probability, secondary signs such as duct dilation) to raise specificity without losing sensitivity.
- Use the vessel and duct labels as extra training classes; pancreatic duct dilation is a known indirect sign of PDAC.
- Evaluate at native resolution and, if possible, on the official PANORAMA test set.

15. References

1 Alves N. et al. PANORAMA: Pancreatic Cancer Diagnosis - Radiologists Meet AI (challenge dataset). Zenodo, 2024. doi:10.5281/zenodo.10599559.

2 Isensee F., Jaeger P. F., Kohl S. A. A., Petersen J., Maier-Hein K. H. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods 18, 203-211 (2021).

3 Isensee F. et al. nnU-Net Revisited: A Call for Rigorous Validation in 3D Medical Image Segmentation. MICCAI 2024.

4 Ronneberger O., Fischer P., Brox T. U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI 2015.

5 He K., Zhang X., Ren S., Sun J. Deep Residual Learning for Image Recognition. CVPR 2016.

6 Alves N. et al. Fully automatic deep learning framework for pancreatic ductal adenocarcinoma detection on computed tomography. Cancers 14(2), 376 (2022).

7 Liu R. et al. An intriguing failing of convolutional neural networks and the CoordConv solution. NeurIPS 2018.